P1 Nov 2010

1.

1. WD = Force × dist
   N m
   Force = ma
   kg ms^{-2}
   WD = ma × dist
   = kg ms^{-2} × m
   = kg m^2s^{-2}

D

2.

2. a = 51 ± 1 cm b = 69 ± 1 cm

A. a + b = 100 ± 2 cm = 2%
B. a − b = 2 ± 2 cm = 100%
C. a × b
add % unc.
= 2% + 2% = 4%
D. $\frac{a}{b}$ add % unc. − 4%

B

3.

3. 2kg → 4N F = ma
$\Rightarrow a = \frac{F}{m} = \frac{4}{2} ms^{-2}$
$a = \frac{v-u}{t}$ $u = 0 \Rightarrow a = \frac{v}{t}$
$v = at = \frac{4}{2} \times 6 = 8 ms^{-1}$

D

4.

4. ½ circumference

av. speed = ½(2π×250)/80

= 5π ms⁻¹ ≈ 16 ms⁻¹



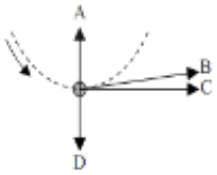
av. vel = 500/50 = 10 ms⁻¹

B

5.

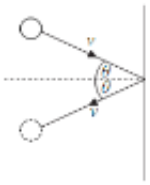
5. Net force is towards the centre

A



6.

6. No change in momt. ⊥ to wall. so only consider ⊥ comp.



Δp = final p − initial p

= −mv cos θ − mv cos θ

= −2mv cos θ

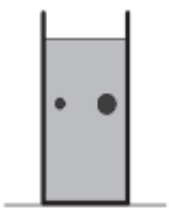
magnitude = 2mv cos θ

D

7.

7. Both both have const. vel. ⇒ net force = zero (same for both)

Bigger ball has larger drag (different)



B

8.

8. If thermally isolated and no work is done ⇒ internal energy must be constant.



A

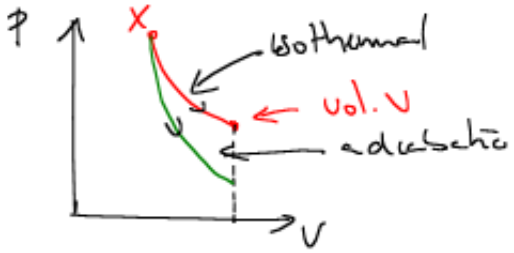
9.

9. Temp same ⇒ Av. KE same
Icebergs larger ⇒ Total KE more

B

10.

10.

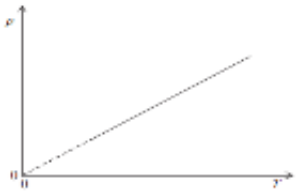


Comparing curves
adiabatic – smaller area
⇒ less work done Less than W
– final temp. is less
⇒ loss of int. energy ΔU – ve
ΔU < 0

D

11.

11. $PV = nRT$
If $P \propto T$
then V must be constant
mass also const ⇒ density constant



A

12.

12.



	x	speed
A	-ve	zero
O	zero	max
B	+ve	zero
O	zero	max
A	-ve	zero

D

13.

13. Period not related to amplitude. In reality (e.g. a pendulum) it is if the ampl. gets big but then it would not be SHM.

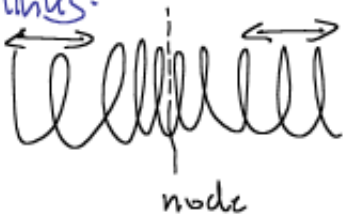
14.

14. velocity of light is less in water but f remains const ⇒ λ reduced.

D

15.

15. Think of a standing longitudinal wave in a slinky.



At the antinodes the spring moves back and forth sometimes compresses the node sometimes stretches it.
⇒ sometimes compression + sometimes rarefaction.

C

16.

16. for 1st minⁿ $d\sin\theta = \lambda$

$\sin\theta = \frac{\lambda}{d}$

so if $\frac{\lambda}{d}$ – greater then pattern is more spread out

$\Rightarrow \frac{\lambda_1}{b_1} < \frac{\lambda_2}{b_2}$

Note, these are not single slit diffraction patterns. – the central max should be brighter and wider than the subsid. max.

17.

17 Sugar solⁿ is optically active → rotates plane of polarisation. B

18.

18. Speed of sound is not changed by moving source but λ is reduced.



C

19.

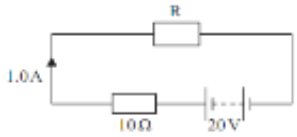
19. $R = \frac{\rho l}{A}$ $\frac{R_x}{R_y} = \frac{\rho l_x}{A_x} \Big/ \frac{\rho b_x/2}{A_x/2} = 1$

C

20.

20. $U = IR$

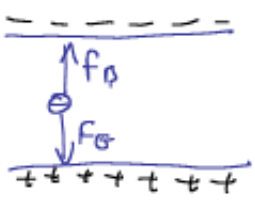
$I = \frac{V}{R} = \frac{20}{10+R} = 1.0$



$\Rightarrow 20 = 10 + R \Rightarrow R = 10\,\Omega$ C

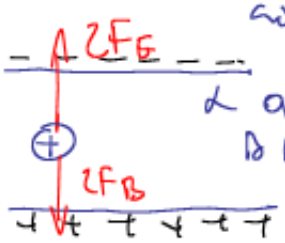
21.

21.



Use Fleming's LHR → B force upwards

If e^- undeflected charge on plates is as shown.



∝ opposite charge ⇒ B force down

$F = Bqv$

q – 2x bigger

⇒ F = 2x bigger than e^-

Charge 2x bigger so F_E 2x bigger than e^-

⇒ No deflection B

22.

22. $g = \frac{GM}{r^2}$ $\frac{g_p}{g_e} = \frac{GM_p}{r_p^2} \Big/ \frac{GM_e}{r_e^2}$

$= \frac{2M_e}{\left(\frac{r_e}{2}\right)^2} \cdot \frac{r_e^2}{M_e} = \frac{2}{(1/2)^2} = 2 \times 4$

$\Rightarrow g_p = 8 g_e$

D

23.

23. Mass does not change the trajectory

D

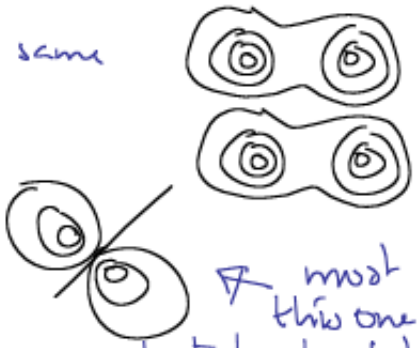
24.

24.

A. two charges same sign

B. masses

C. dipole



∴ most like this one

Lines of equipotential close between charges ⇒ large field.

25.

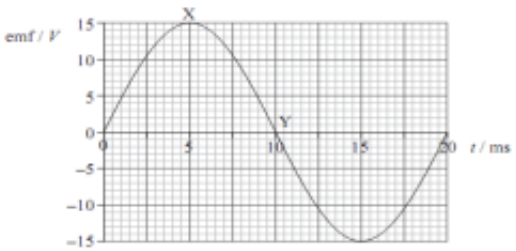
25. 

Along axis distance from B = dist from D & Dist from A = dist from C

⇒ B

26.

26. 



Current ∝ V ⇒ max^m at X

EMF ∝ rate of change of flux linkage

When flux linkage is max^m there is little change so no EMF induced

⇒ flux linkage max at Y B.

27.

27. $P = \frac{V_{rms}^2}{R} = \left(\frac{15/\sqrt{2}}{5}\right)^2 = 22.5 = \frac{45}{2}\,W$

A.

28.

28. 16 → 8 → 4 → 2 → 1
(½ ½ ½ ½)

⇒ 4 ½ lives
If ½ life = 2 minutes

⇒ time take = 8 mins B.

29.

29. Geiger and Marsden showed that atom had a small +ve nucleus.
C.

30.

30. Nuclear radius can be determined by scattering experiments A.

31.

31.

KE increases if f increases rather than increased amplitude. This can be explained if light is made of photons each of energy hf. This can not be explained by wave theory.

If light was a wave then low intensity light would be low amplitude. It would take a long time for an atom to gain enough energy to eject an electron. The lack of time delay can not be explained by the wave model

⇒ Neither can be explained A.

32.

32. $\lambda = \frac{h}{p} = \frac{h}{mv}$ same λ so

$\frac{h}{m_p v_p} = \frac{h}{m_\alpha v_\alpha} \Rightarrow m_p v_p = m_\alpha v_\alpha$

$m_\alpha = 4 m_p \Rightarrow m_p v_p = 4 m_p v_\alpha$

$\frac{v_\alpha}{v_p} = \frac{1}{4}$ A.

33.

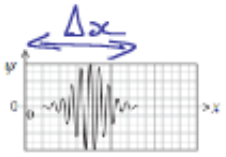
33. Discrete α energy $\Rightarrow$ energy level structure in the nucleus B.

34.

34. $\Delta x \Delta p \geq \frac{h}{4\pi}$

$\Rightarrow$ if Δx small then Δp big

C has the smallest Δx

C. 

C.

35.

35. Power $= \frac{\rho v S A^2}{2}$

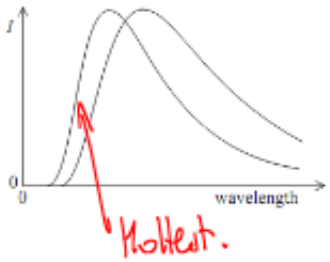
$\Rightarrow P \propto A^2 v$ $\frac{4}{P} = \frac{A^2 v}{(2A)^2 2v} = \frac{1}{8}$

$\Rightarrow P = 32\ kWm^{-1}$ C.

36.

36.

Temp ∝ $1/\lambda_{max}$
so don't have same temp



Curves are neither perfect black bodies
⇒ different emissivity.

37.

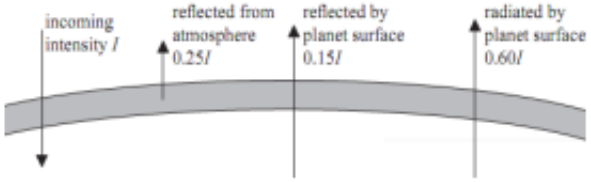
37. Nuclear fuel does have a high energy density

A.

38.

38. $\text{albedo} = \frac{\text{reflected}}{\text{incident}} = \frac{0.25I + 0.15I}{I}$

$= 0.4$ C.



39.

39.

f samples/s ⇒ each sample takes $\frac{1}{f}$ s

Total number of bits = N

each sample contains b bits so no. of samples = $\frac{N}{b}$

so total time = $\frac{N}{bf}$ B.

Confusing

40.

40. Charge on pixel is related to number of electrons which is related to number of photons which in turn is related to the intensity.

B

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